# Schema inference and entity identification

One runtime agent generated six mapping configs for a generic extractor, producing 22,925 observations, 225 linked groups and 50 company profiles. This is a partial submission. Saved outputs use runtime `e3061c95`; the current code, `9aa20f50`, contains later experimental changes. Those experiments have not replaced the saved cohort or established reliable generalisation.

## 1. Cost and time

The author's latest API spend estimate is **US$3.10**. This is not a reconciled project total: an earlier archived self-report was US$6-7, with unresolved scope differences. The saved six-source run has provider receipts for **87 calls, 1,563,821 tokens and US$0.2054033**, taking **262.234 seconds** with three workers. Companies alone used **14 calls, 383,438 tokens, US$0.0524642 and 178.651 seconds**, including acquisition. Its inference trace records seven events, including two candidate probes, two name reviews and development validation. Events and model calls are different counts.

Inference is an onboarding cost per source/config version. AST extraction makes **zero per-record model calls**; CPU, storage and human review costs are separate. I estimate **7-10 hours of cumulative active effort**: 3-4 on framework design, 3-4 on experiments, debugging and fixes, and 1-2 exploring generalisation. This updates an earlier under-six-hours active-time estimate and exceeds the six-hour time box; it is not elapsed calendar time.

## 2. Agent design and self correction

Deterministic catalogue filtering ranked 50 candidates from 1,290 records, avoiding model calls for obvious exclusions. A fixed random sample of 20 received human-confirmed, AI-assisted review: 19 contained identifiable businesses and one remained uncertain. The possible false-positive fraction within this sample is 0-5%, not a population estimate.

The agent inspects native tables and documents, samples records, proposes mappings and executable expressions, probes them, validates on initially withheld development records, and revises or selects a candidate. Separate steps expose execution and evidence failures while carrying state forward. For Companies, it revised the name proposal to use the companion Current Name field on historical-name rows. Unsupported fields remain unmapped with reasons. Operator-declared required fields guided task acceptance; autonomous requirement discovery is unproven.

Human review exposes mappings, quotations, examples, omissions and unresolved requirements. Approval binds config, evidence, ontology and runtime. All approvals, including the saved cohort, were synthetic; no config has human approval. Persisted requests, artifacts and checkpoints allow completed steps to be recovered. Failed document retrieval preserves local inputs; unknown paid outcomes stop without automatic resend.

The weakest step is table/subject interpretation and field discovery. In later eight-source runs, Companies type coverage fell from 386 to zero, while 693 APRA identifiers switched from trustee to fund. More emitted fields did not establish better semantic quality. Source claim time is also unresolved: all saved observations use labelled ingestion or dataset-update proxies, excluded from recency comparisons.

## 3. The difficult source

ATO combines business tables and workbook explanations. The saved run retained 4,436 business rows; an earlier attempt included six explanatory rows. Its generic Name field lacks registered-name evidence, so legal_name remains unmapped. I would require explicit table dispositions and subject definitions. A custom adapter is justified for a documented format gap when a reusable reader is uneconomical; it should be isolated, tested and disclosed, without inventing field semantics.

Later holdouts exposed reader limits: CRICOS failed on merged cells before inference; Aged Care generated address mappings but read single-space cells as empty strings in 10,423 cells across 5,377 rows. Address-component agreement does not make that end-to-end result acceptable.

## 4. Entity relationships and profiles

Qualified shared typed identifiers produce candidate pairs and entity groups. Subject/name conflicts and ambiguous identifiers produce abstentions. The saved build has **448 cross-source pairs and 22,278 unlinked records**: 132 conflict/ambiguity refusals and 22,146 without a qualified identifier or cross-source anchor. Names alone never confirm a link. Parent/subsidiary and brand/holder relationships are not identity; without source evidence, I leave them unasserted.

The fixed **0.99 threshold is an uncalibrated policy score**. The saved cohort's 50-link sample has no human decisions (0/50). A separate revised-policy audit contains 18 human and 32 AI judgments; it cannot establish precision for the saved cohort. Measured human precision remains unavailable.

Profiles carry field confidence, provenance and selection reasons; missing values are omitted. Conflict retention is implemented, but none of the submitted 50 profiles has a conflicting field, so real-conflict display is not demonstrated. Removing Companies affects **44 of the submitted 50 profiles**, losing **166 field values**. Withdrawal retains membership and flags identity review, rather than proving the remaining clusters.

## 5. Scale from six sources to 500

The human review queue is likely to constrain onboarding first: unreliable documents and ambiguous definitions consume repeated calls and human attention. Dense identifier groups also create pairwise work. I would cache versioned evidence, partition candidates by typed identifiers, and measure review time, useful coverage and cost per accepted source. These are scaling hypotheses, not benchmarks.

## 6. Replacing the matching model

Before changing millions of links, retain immutable observations, native locators, evidence/config/model/ontology hashes, matcher policy versions, link reasons and human decisions. Keep proposed, accepted and rejected relationships distinguishable, so a new matcher can reconsider decisions without overwriting history.

Run old and new matchers on the same frozen observations. Compare added/removed pairs and cluster splits/merges, trace their affected profiles, and review risky differences. Include previously rejected records to expose recall changes, rather than evaluating accepted links alone. Stage activation by bounded partitions, monitoring false merges and review volume.

Version memberships and derived profiles together. A rollback must restore both, with an explicit mapping between old and new entity keys. Existing withdrawal recomputation is narrower than this migration plan; deployment and rollback have not been demonstrated.

## 7. Next three weeks

First fix raw-value preservation and table/subject discovery. Then complete the human audit for the chosen final matcher and measure field omissions, false accepts and review cost on fresh sources. Finally test versioned matcher migration. Keep field confidence separate from link confidence, retain useful source observations, and prioritise reliable review over adding more source-specific features.